A FRAUD-MODEL POST-MORTEM — BEFORE DEPLOYMENT

My fraud model hit PR-AUC 0.87.

The validation report still filed 8 findings against it.

I built the model. Then I switched roles and challenged it the way a bank's Model Risk team would.

Swipe for what the metrics were hiding →

WHY BOTHER?

Banks run three lines of defence. Portfolios only ever show Line 1.

Line 1 builds the model. Line 2 — Model Risk — independently proves whether it's fit for purpose. Regulators have required this since 2011 (SR 11-7), and still do under its 2026 replacement, SR 26-2.

So this project wears both hats, honestly: same model, same data, a wall between the roles.

HAT 1 — THE DEVELOPER

A production-shaped application-fraud model

80k applications · fraud rings sharing devices & addresses · point-in-time features · time-ordered split · calibrated LightGBM.

0.873

PR-AUC on the out-of-time test fold

99.2%

precision in the top-1% investigation queue

Looks shippable. Most projects stop here.

HAT 2 — THE VALIDATOR

Then I tried to break my own model.

The independent validation suite:

- Challenger models — must the champion be this complex?
- Feature-family ablation — any single point of failure?
- Input perturbation — can an applicant steer the score?
- Label-quality scan (confident learning)
- PSI drift analysis across time windows
- SHAP review — would explanations survive challenge?
- Segment analysis — who does it miss?

FINDING 1 · HIGH

A simpler model beat the champion.

0.887 vs 0.873

challenger PR-AUC vs champion — a 300-tree shallow GBM outperformed the tuned 2000-tree model

The champion's complexity was unearned. Nobody checks this unless a challenger benchmark forces the question.

FINDING 2 · HIGH

My "multi-factor" model was secretly a one-trick model.

Remove all 11 entity-linkage features: PR-AUC barely moves (–0.004).

0.873 → 0.466

PR-AUC when the 4 affordability ratios are removed

The stated design theory and the model's actual behaviour were two different things. Ablation is how you find out.

FINDING 3 · HIGH

An applicant could steer the queue.

92%

of the top-1% investigation queue turns over if applicants understate income by 20%

The dominant feature is self-reported and unverified. That's not a model bug — it's an attack surface. Condition of approval: income verification for high-score applications.

FINDINGS 4–8

And the quieter ones:

- Segment blind spot — 90% of ring fraud caught, only 57% of solo fraud. Headline metrics averaged it away.
- Drift-alarm fatigue — 9 of 24 features breach PSI thresholds while the score is perfectly stable. Naive monitoring would cry wolf daily.
- Age-proxy risk — age carries high attribution; fine for triage, a discrimination problem the day scores gate decisions.

THE VERDICT

Fit for purpose — with conditions.

Approved for human-reviewed triage only, conditional on fixing the three High findings and a monitoring plan with governance-calibrated thresholds.

Same model, different intended use — auto-decline instead of triage — and the verdict flips to NOT FIT. Purpose is part of the model.

THE POINT

7 of these 8 findings are invisible to train → test → ship.

They only surface when someone whose job is NOT shipping the model goes looking. That's what AI validation is — and why it's a discipline, not a checkbox.

Everything here is reproducible: synthetic data with known ground truth, single-command pipeline, full validation report on my GitHub (Kriti-Data-Business).

Next: validating an LLM fraud-triage assistant — the part current regulation doesn't even cover yet. Follow along.